

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	XXXXXXXXXX
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution)

The proposed solution is to develop **Rising Waters**, a machine learning-based web application that predicts flood risk using historical weather and rainfall data. The system analyzes environmental parameters such as temperature, humidity, cloud cover, and rainfall to determine whether flood conditions are likely. By providing timely flood risk predictions through a simple and user-friendly Flask web interface, the application supports early awareness and preparedness. The solution aims to demonstrate the practical use of machine learning in flood prediction and help users make informed decisions based on weather-related data

Project Overview	
Objective	Develop a machine learning-based web application to predict flood risk using historical weather and rainfall data.
Scope	The system analyzes weather parameters to predict flood risk and provides users with instant results through a Flask web application.
Problem Statement	
Description	Floods cause significant damage to lives and property, making early prediction essential. Manual forecasting methods are often time-consuming and less efficient.
Impact	The proposed system supports early flood risk assessment, improves preparedness, and helps reduce the potential impact of flood-related disasters.
Proposed Solution	
Approach	Develop a machine learning-based web application that analyzes historical weather and rainfall data to predict flood risk. The system uses a trained Random Forest model integrated with a Flask web application to provide instant flood predictions.
Key Features	• Predicts flood risk using machine learning. • Provides a simple and user-friendly web interface. • Displays prediction results instantly based on user inputs. • Includes Analytics and Contact pages for model information and emergency helpline numbers.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
	Computing Resources	Intel Core i5/i7 processor (or equivalent)
	Memory	Minimum 8 GB RAM
Storage	Storage	Minimum 256 GB SSD
Software		
	Frameworks	Flask – Web application framework
	Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle
	Development Environment	Jupyter Notebook, Visual Studio Code
Data		
Data	Source, size, format	Kaggle Flood Prediction Dataset, CSV format (115 records, 10 features)